# FEDOSOV QUANTIZATION OF POISSON MANIFOLDS AND QUANTUM MOMENT MAPS

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Abstract. We construct invariant star products with quantum moment maps on Poisson manifolds using a Lagrangian embedding into a formal symplectic groupoid and performing Fedosov quantization, providing an explicit realization for the existence theorem of quantum moment maps on Poisson manifolds recently proven by Esposito-Nest-Schnitzer-Tsygan via $\mathfrak{g}$-adapted formality. We find that the weaker condition of a $G$-invariant connection is sufficient for its existence. Conversely, given a bi-Lagrangian foliated symplectic manifold with a $G$-action, we induce a non-trivial Poisson structure and its deformation on a Lagrangian foliation, which then admits projected quantum moment maps for suitable $G$.

## 1. Introduction

In this paper we construct a Fedosov star product on a Poisson manifold $M$ by embedding it into its integrating formal symplectic groupoid $(X, \omega)$, performing the Fedosov construction on $X$, and descending the star product to $M$. When $M$ carries a Hamiltonian $G$-action and a $G$-invariant connection, the descended star product is $G$-invariant and admits quantum moment maps given by explicit formulas. This provides a concrete realization of the existence results recently obtained by Esposito-Nest-Schnitzer-Tsygan [4] through $\mathfrak{g}$-adapted formality, under the weaker assumption of the existence of a $G$-invariant connection on $M$, as compared to the equivariant projectivity condition in [4] for the $\mathfrak{g}$-adapted formality morphism.

The formal symplectic groupoid, constructed by Cattaneo-Dherin-Felder [2], extends the cotangent Lie algebroid integration of Weinstein by carrying the Poisson structure of $M$ into a symplectic structure on a formal neighborhood $X$. We exploit the bi-Lagrangian splitting of $(X, \omega)$ to decompose the Fedosov connection into normal and tangential components, and show that the kernel of the normal component forms a subalgebra of the Weyl bundle isomorphic to a Weyl bundle on $M$. The star product on $M$ is then defined by descending the Fedosov flat sections through this isomorphism. For the quantum moment map, we lift the classical moment map from $M$ to $X$ via the same mechanism, correct it by a Chevalley-Eilenberg cocycle when necessary, and project back. The obstruction to this correction is a class in $H^2_{\mathrm{CE}}(\mathfrak{g}, \mathbb{C}[[h]])$ which vanishes when $\Omega = 0$, and for any $\Omega$ when $\mathfrak{g}$ is semisimple, where $\Omega$ is the Fedosov central curvature. Following are the main results we obtain.

**Theorem A.** *Let $(M, \pi)$ be a Poisson manifold with a Hamiltonian $G$ action (with Lie algebra $\mathfrak{g}$) and classical moment maps $\mu$. Assume $M$ admits a $G$-invariant connection. Then there exists a $G$-invariant star product $\star_M$ on a formal neighbourhood of $M$ with quantum moment maps $\hat{\mu}$ satisfying*

$$[\hat{\mu}_\xi, \hat{\mu}_\eta]_{\star_M} = ih\hat{\mu}_{[\xi,\eta]} \tag{1}$$

*for all $\xi, \eta \in \mathfrak{g}$, and is given by an explicit Fedosov construction on $M$ with central curvature $\Omega = 0$*

**Theorem B.** *If $\mathfrak{g}$ is semisimple, then every such $G$-invariant Fedosov star product admits a unique quantum moment map $\hat{\mu}_{\mathfrak{g}} = \hat{\mu} - a$ satisfying* (1) *where $a \in \mathfrak{g}^* \otimes \mathbb{C}[[h]]$ depends only on $\Omega$*

## 2. Embedding of a formal neighborhood

We view the Poisson manifold $(M^n, \pi)$ as being embedded inside a $2n$ dimensional manifold $X$, locally defined using constraint equations $M := \{x \in X \mid x^\alpha = 0, \alpha = 1 \cdots n\}$ and the Poisson bivector of $M$ with coefficients $\pi^{ij}(x)$, $x \in M$. Here, the latin superscripts denote coordinates of $M$, and the greek ones denote coordinates of $X$ which are normal to $M$. We define a bi-Lagrangian symplectic structure on $X$ as follows.
Let $\Gamma^i_\alpha$ and $\omega^{i\alpha}$ be functions on $X$ which satisfy :

$$\pi^{ij} = \omega^{j\alpha}\Gamma^i_\alpha - \omega^{i\alpha}\Gamma^j_\alpha \tag{2}$$

$$\mathrm{Rank}(\omega^{i\alpha}) = n \tag{3}$$

for all $\alpha, \beta, i, j$. $\omega$ is then a bi-Lagrangian symplectic form, after we let $d\omega = \omega^{ij} = \omega^{\alpha\beta} = 0$ Moreover, we construct a partial connection in the normal directions

$$D = dx^\alpha \otimes D_\alpha = dx^\alpha \otimes (\partial_\alpha + \Gamma^i_\alpha \partial_i + x^\beta A^i_{\alpha\beta}\partial_i + x^\beta x^\gamma A^i_{\alpha\beta\gamma}\partial_i \cdots) \tag{4}$$

where the coefficients $A^i_{\alpha\beta\ldots}$ are chosen iteratively to ensure flatness of $D$. For example, the first order coefficient is

$$A^i_{\alpha\beta} = -\left(\partial_\alpha(\Gamma^i_\beta) + \Gamma^j_\alpha \partial_j(\Gamma^i_\beta)\right)$$

**Lemma 2.1.** *Flat sections of $D$ correspond to functions on $M$*

*Proof.* We prove this by constructing an explicit lifting map $\sigma^{-1} : C^\infty(M) \longrightarrow C^\infty(X)$ with the reverse direction being the projection $\sigma$ defined as restriction to $x^\alpha = 0$. Let $f(x^i) \in C^\infty(M)$ and $F(x^i, x^\alpha) = \sigma^{-1}(f)$. Write $F$ formally as a taylor series in normal variables

$$F(x^\alpha, x^i) = f + f_\alpha x^\alpha + f_{\alpha\beta} x^\alpha x^\beta + \cdots \tag{5}$$

where the coefficients $f_{\alpha\beta\ldots}$ are functions on $M$. We can find each coefficient by solving $D(F) = 0$ iteratively in powers of $x^\alpha$.

□

**Proposition 2.2.** *There always exists a solution for $(\omega, \Gamma)$ satisfying the above conditions, and a lifting map $\sigma^{-1}$ which satisfies*

$$\{f, g\}_M = \sigma\{\sigma^{-1}f, \sigma^{-1}g\} = \{F, G\}|_M \tag{6}$$

*where $\{\,,\,\}$ and $\{\,,\,\}_M$ are the Poisson brackets with components $\omega$ and $\pi$ respectively.*

*Proof.* The pair $\omega^{i\alpha} = \delta^{i\alpha}, \Gamma^i_\alpha = \delta_{j\alpha}\pi^{ij}/2$ satisfies (2) and (3) as the trivial solution. The flatness condition $D^2 = 0$ determines the higher coefficients $A^i_{\alpha\beta\ldots}$ at each order, and uniqueness of the lifting map $\sigma^{-1}$ for a given $D$ follows from Lemma 2.1 □

**Remark 2.3.** We may identify the lifting map $\sigma^{-1}$ with the pullback of the deformed source map $s$ in [2] which identifies $M$ with the "source" Lagrangian submanifold of the formal symplectic groupoid $X$.

**Remark 2.4.** One may globalize this trivial solution and the rest of the following construction by using the jet bundle approach described in [3]. We allow a more general $(\omega, \Gamma)$ since this may be relevant when $M$ is a genuine Lagrangian submanifold of a physical symplectic manifold $(X, \omega)$ where the embedding is fixed by the ambient setup rather than chosen freely.

## 3. Fedosov quantization and descent to $M$

With $(X, \omega)$ as a symplectic manifold, we can get a star product using Fedosov's construction [5]. We describe the procedure here in brief and establish notation.

3.1. **The Weyl bundle and Fedosov connection.** Fedosov defines a canonical noncommutative algebra over functions $C^\infty(X)[[h]]$ of $X$ by using the Weyl bundle $\mathcal{W}$ consisting of the $C^\infty(X)[[h]]$ module generated by formal tangent variables $(y^1, y^2 \dots) \in TX$.
The Weyl bundle is given an algebra structure with the Moyal product defined as :

$$\begin{aligned}(a \circ b) &= \exp\left(-\frac{ih}{2}\omega^{IJ}\frac{\partial}{\partial y^I}\frac{\partial}{\partial z^J}\right) a(y,h)b(z,h)|_{z=y} \\ &= \sum_{k=0}^{\infty}\left(-\frac{ih}{2}\right)^k \frac{1}{k!}\omega^{i_1,j_1}\cdots\omega^{i_k j_k}\frac{\partial^k a}{\partial y^{i_1}\cdots\partial y^{i_k}}\frac{\partial^k b}{\partial y^{j_1}\cdots\partial y^{j_k}}\end{aligned} \tag{7}$$

We lift $C^\infty(X)[[h]]$ to sections of $\mathcal{W}$ by using a connection that preserves the Moyal product structure across different charts. It is defined as

$$D_{\text{Fed}} = -\delta + \partial + \left[\frac{i}{h}r, \cdot\right]_\circ \tag{8}$$

With the Koszul differential $\delta = dx^I \dfrac{\partial}{\partial y^J}$ and base symplectic connection $\partial$. The correction term $r$ starting with cubic powers of $y$ is chosen to cancel the curvature of the symplectic connection and thus make $D_{\text{Fed}}$ flat. Flat sections of $D_{\text{Fed}}$ are in bijection with $C^\infty(X)[[h]]$ via the Fedosov lifting map $\sigma_{\text{Fed}}^{-1} : C^\infty(X)[[h]] \longrightarrow \Gamma(\mathcal{W})$, with its inverse $\sigma_{\text{Fed}}$ being restriction to $y^i = 0$ and

$$D_{\text{Fed}}(\sigma_{\text{Fed}}^{-1}F) = 0 \text{ for all } F \in C^\infty(X)[[h]]$$

3.2. **Tangent - normal decomposition.** We split the tangent bundle of $X$ at $M$ into its tangent and normal components, allowed due to the bi-Lagrangian structure

$$T_M X = TM \oplus N_M$$

where $N_M$ is the normal distribution generated by $\partial_\alpha$ restricted to $M$. We can decompose the Fedosov connection along these vector bundles

$$D_{\text{Fed}} := D^\perp + D^\parallel = dx^\alpha D^\perp_\alpha + dx^i D^\parallel_i \tag{9}$$

3.3. **The $D^\perp$-flat subalgebra.**

**Lemma 3.1.** ${D^\perp}^2 = 0$ *and* $D^\perp$ *is a derivation of* $\circ$

*Proof.* Since $TX$ splits along $M$, the identity $D^2_{\text{Fed}} = 0$ can be decomposed into $dx^i \wedge dx^j$, $dx^i \wedge dx^\alpha$, and $dx^\alpha \wedge dx^\beta$ components, each of which have to vanish. Similarly, the distributive property of $\circ$ proves the second half. □

**Proposition 3.2.** $\mathcal{W}^\perp := \ker D^\perp$ *is a subalgebra of the Weyl algebra* $\mathcal{W}$

This follows from the previous lemma

3.4. **The star product on $M$.** Let $\mathcal{W}_M = (C^\infty(M)[[h]])[[y^i]]$ be the Weyl bundle restricted to $M$ in both its module and algebra structures. The sections $\Gamma(\mathcal{W}_M)$ of this bundle are power series in $y^i$ with coefficients in $C^\infty(M)[[h]]$. This is a subalgebra, and the original Moyal product acts trivially, since $\omega^{ij} = 0$. We now induce a different product on this subalgebra using a lift similar to the classical one, such that the image lands in $\mathcal{W}^\perp$. Define

$$\begin{aligned} \Phi : \Gamma(\mathcal{W}_M) &\longrightarrow \Gamma(\mathcal{W}^\perp) \subset \Gamma(\mathcal{W}) \\ s(x^i, y^i) &\longmapsto \tilde{s}(x^i, x^\alpha, y^i, y^\alpha) \end{aligned} \tag{10}$$

where $\Phi(s)$ is defined as the unique $D^\perp$ flat section that restricts to $s$ on $\mathcal{W}_M$

**Theorem 3.3.** *$\Phi$ is an algebra isomorphism with inverse being projection to $\mathcal{W}_M$ and the product on $\Gamma(\mathcal{W}_M)$ defined as*

$$s_1 \circ_M s_2 = (\tilde{s}_1 \circ \tilde{s}_2)|_{y^\alpha, x^\alpha = 0}$$

*Proof.* We prove this using a similar argument as Theorem 3.3 in [5]. Let $\delta_n$ be the Koszul operator present in $D^\perp$ and let $\delta_n^{-1}$ be its inverse

$$\delta_n = dx^\alpha \frac{\partial}{\partial y^\alpha} \qquad\qquad \delta_n^{-1} = \frac{1}{p+q} y^\alpha \iota_{\frac{\partial}{\partial x^\alpha}}$$

when acting on a homogenous element $s \in \mathcal{W} \otimes \Sigma$ of $y^\alpha$ degree $p$ and $dx^\alpha$ degree $q$. These satisfy the following identity

$$s = \delta\delta^{-1}s + \delta^{-1}\delta s + s_{00} \tag{11}$$

where $s_{00}$ is the restriction of $s$ to $\mathcal{W}_M$.
Let $\tilde{s} \in \mathcal{W}^\perp = \mathrm{Ker}(D^\perp)$. Using the definition of $D^\perp$, this is

$$\delta_n \tilde{s} = \partial_n \tilde{s} + \left[\frac{i}{h} r_n, \tilde{s}\right] := A(\tilde{s})$$

where $\partial_n = dx^\alpha \partial_\alpha$ are the normal derivatives, and similarly $r_n$. Applying $\delta_n^{-1}$ to both sides and using (11), this becomes

$$\tilde{s} = s_{00} + \delta_n^{-1}(A(\tilde{s})) \tag{12}$$

Since $\delta_n^{-1}$ necessarily adds a $y^\alpha$ term to $A$, we see that $\Phi^{-1}\tilde{s} = s_{00}$. Thus, $\mathcal{W}^\perp \subset \mathrm{Im}(\Phi)$. Conversely, let $\tilde{s}$ be a section of $\mathcal{W}$ which satisfies (12). We need to show that $D^\perp \tilde{s} = 0$. Define

$$B = D^\perp \tilde{s} = -\delta_n \tilde{s} + A(\tilde{s}) \tag{13}$$

Since ${D^\perp}^2 = 0$, $D^\perp B = 0$; and we have

$$\delta_n B = A(B) \tag{14}$$

Apply $\delta_n^{-1}$ on (13) to get

$$\delta_n^{-1} B = -\delta_n^{-1}\delta_n \tilde{s} + \delta_n^{-1} A(\tilde{s}) = -\tilde{s} + s_{00} + \delta_n^{-1} A(\tilde{s}) = 0$$

Finally, applying $\delta_n^{-1}$ to (14) and using the fact that $B$ is a 1-form, thus $B_{00} = 0$ and we get

$$B = \delta_n^{-1} A(B)$$

Because $\delta_n^{-1}$ increases the $y$ degree, solving this equation iteratively implies that $B = 0$.
Additionally, $D^\perp$ is a derivation of $\circ$, thus the Moyal product is closed inside $\mathcal{W}^\perp$. This

combined with the fact that $\Phi$ is an isomorphism proves that it is also an algebra homomorphism

$\square$

We can now construct the Fedosov flat connection directly on $M$ using $\Phi$. Define

$$\bar{D} := \Phi^{-1} D^{\parallel} \Phi \tag{15}$$

**Lemma 3.4.** *$\bar{D}$ is flat, and a derivation of $\circ_M$*

*Proof.* This follows from flatness of $D^{\parallel}$ and Theorem 3.3 $\square$

We can repeat the same arguments to show that there is an isomorphism between $C^\infty(M)[[h]]$ and flat sections of $\bar{D}$ in $\mathcal{W}_M$. We call this map $\bar{\sigma}^{-1}$, and descend the algebra structure via $\bar{\sigma}$

**Theorem 3.5** (Star product on a Poisson manifold)**.** *The star product on $C^\infty(M)[[h]]$ defined as*

$$f \star_M g := \bar{\sigma}(\bar{\sigma}^{-1}(f) \circ_M \bar{\sigma}^{-1}(g))$$

*is well defined and associative*

*Proof.* The result follows from the fact that $\bar{\sigma}$ is an algebra isomorphism. $\square$

3.5. **Classical limit.** Let $f \in C^\infty(M)[[h]]$. Let $\bar{\sigma}^{-1} f$ be its lift to $\Gamma(\mathcal{W}_M)$ that satisfies $\bar{D}\bar{\sigma}^{-1} f = 0$.

**Proposition 3.6.** *$\Phi(\bar{\sigma}^{-1} f)$ is a flat section of $(\mathcal{W}, D_{\mathrm{Fed}})$*

*Proof.* Since $\bar{D}\bar{\sigma}^{-1} f = 0 \implies \Phi^{-1} D^{\parallel}(\Phi\bar{\sigma}^{-1} f) = 0$
Thus, $\Phi(\bar{\sigma}^{-1} f)$ is $D^{\parallel}$ flat, and also $D^{\perp}$ flat by definition. $\square$

**Theorem 3.7.** *$\star_M$ satisfies the correspondence principle*

$$f \star_M g = fg + O(h), \qquad [f, g]_{\star_M} = ih\{f, g\}_M + O(h^2)$$

*Proof.* Because every flat section of $D_{Fed}$ is the lift of a function on $X$,

$$\Phi(\bar{\sigma}^{-1} f) \circ \Phi(\bar{\sigma}^{-1} g) = \sigma_{\mathrm{Fed}}^{-1}(F) \circ \sigma_{\mathrm{Fed}}^{-1}(G) = \sigma_{\mathrm{Fed}}^{-1}(F \star_X G) \tag{16}$$

where $F, G \in C^\infty(X)[[h]]$ are such that $\Phi(\bar{\sigma}^{-1} f) = \sigma_{\mathrm{Fed}}^{-1}(F)$ and $\Phi(\bar{\sigma}^{-1} g) = \sigma_{\mathrm{Fed}}^{-1}(G)$.
Applying $\bar{\sigma}\Phi^{-1}$ on both sides, we get

$$f \star_M g = (F \star_X G)|_M \tag{17}$$

Taking the $h = 0$ limit on both sides, we find that $F|_M = f$ and similarly for $g$. For an arbitrary choice of the symplectic connection, these $F, G$ might not match the lifts $\sigma^{-1} f$ and $\sigma^{-1} g$ we had constructed in (5). However, this is not a problem, since in [5] it is also proven that Fedosov star products are classified upto equivalence by

$$\frac{1}{h}[\omega] + H^2_{\mathrm{dR}}(M, \mathbb{C})[[h]]$$

they depend only on the choice of the closed 2-form $\Omega \in H^2_{\mathrm{dR}}(M, \mathbb{C})[[h]]$ which turns out to be the central curvature term left in $D_{\mathrm{Fed}}$, and it is of the form

$$\Omega = h\Omega_1 + h^2\Omega_2 \cdots$$

where each $\Omega_i$ is closed. This means that for fixed $\omega$ and $\Omega$ any two star products defined using connections $\nabla$ and $\nabla'$, the antisymmetric part of the first order bidifferential operator must match the Poisson bracket of $\omega$

$$F \star_X G - G \star_X F = ih\{F, G\} + O(h^2) \tag{18}$$

$\square$

**Proposition 3.8.** *The classical limit of $[f, g]_{\star_M}$ is also invariant under choice of symplectic connection, and reproduces the required Poisson bracket on $M$*

$$f \star_M g - g \star_M f = ih\{F, G\}|_M = ih(\pi^{ij}\partial_i f \partial_j g) \tag{19}$$

This follows because (19) is simply a restriction of (18). To demonstrate this locally in the trivial Darboux coordinates, we can select a particular choice of $\nabla_X$ adapted to the lift $\sigma$ such that the classical lift $\sigma^{-1}f$ matches $F$. Define the Christoffel symbols

$$\Gamma^i_{\alpha j} = \Gamma^i_{j\alpha} = \partial_j \Gamma^i_\alpha \qquad \Gamma^\alpha_{ji} = -\partial_j \Gamma^i_\alpha \tag{20}$$

with all other components zero. This is torsion-free, with zero curvature, and $\star_M$ with this connection reduces to the standard Moyal product. It is $G$-invariant if $\pi$ is, but only works locally. To make sure the product behaves well under coordinate changes, we must use the $G$-invariant connection on $M$ to define $\nabla_X$

## 4. $G$-invariance and the cotangent lift

Let $\rho : \mathfrak{g} \to C^\infty(M)$ be a classical moment map for the Hamiltonian $G$-action on $(M, \pi)$:

$$\{\rho(\xi),\, \rho(\eta)\}_M = \rho([\xi, \eta]) \qquad X^M_\xi = \{\rho(\xi), \cdot\}_M \qquad \text{for all } \xi, \eta \in \mathfrak{g} \tag{21}$$

Since $X$ is formally the symmetric tensor algebra of the cotangent bundle, the $G$-action induces a canonical action on $X$ which preserves $M$ as the zero section. Moreover, the canonical symplectic form $\omega$ is also invariant under this induced action. A $G$-invariant connection on $M$, which we may assume torsion-free without loss of generality by symmetrizing, canonically induces a $G$-invariant symplectic connection $\nabla_X$ on $X$ compatible with $\omega$: the torsion-free condition ensures that the induced horizontal distribution on $T^*M$ is Lagrangian, and $G$-invariance transfers directly since the construction is functorial. From here on we work with this connection.

**Remark 4.1.** Note that we do not need to restrict ourselves to this induced $G$-action. The only requirement is that $\omega$ and $\Gamma$ be $G$ invariant, the lift $\sigma^{-1}$ (or deformed source map) which relates the two Poisson structures be $G$-equivariant. If this holds, any flow $X_\xi$ generated by a moment map $J_\xi$ on $X$ as

$$X_\xi = \{J_\xi, \cdot\}$$

that does not preserve $M$ still projects via $\sigma$ to $M$ which is a genuine symmetry action due to (6). In that case we call $\rho_\xi = \sigma(J_\xi)$ a hidden symmetry, since it does not emerge from a reduction of the ambient Poisson structure, but from a projection of it. Of course, as long as $\omega^{\alpha\beta} = 0$ we may never get this, but we leave the intermediate $\omega^{\alpha\beta} \neq 0$ case for a future project.

Given a valid $G$-action on $X$ and a $G$-invariant symplectic form, all the ingredients for the rest of the quantization procedure automatically become $G$-invariant, since there is no other $G$-dependent input in the construction except $\Omega$, which we are free to choose.

**Proposition 4.2.** *Let $(M, \pi)$ be a Poisson manifold with a Hamiltonian $G$-action, and suppose $M$ admits a $G$-invariant connection. Then for any $G$-invariant $\Omega$, the star product $\star_M$ constructed in Theorem 3.5 is $G$-invariant:*

$$\phi_g^*(f \star_M h) = \phi_g^* f \star_M \phi_g^* h \qquad \forall\, g \in G,\ f, h \in C^\infty(M)[[h]] \tag{22}$$

*Proof.* $\nabla_X$ is $G$-invariant by construction and $\Omega$ is $G$-invariant by assumption. By Theorem 3.8 in [8] $\star_X$ is $G$-invariant. Since $\sigma_{\mathrm{Fed}}, \Phi, \bar\sigma$ are all $G$-invariant, the descended product is also $G$-invariant. □

## 5. Quantum moment maps

5.1. **Lift of the classical moment map.** For each $\xi \in \mathfrak{g}$, the element $\Phi(\bar\sigma^{-1}(\rho(\xi)))$ lies in $\ker D_{\mathrm{Fed}} \cap \mathcal{W}^\perp$ by the argument of Theorem 3.3. Define $J(\xi) \in C^\infty(X)[[h]]$ by

$$\sigma_X^{-1}(J(\xi)) = \Phi(\bar\sigma^{-1}(\rho(\xi))) \tag{23}$$

Then $J(\xi)\big|_M = \rho(\xi)$. This is the lifted moment map to $X$.
[8] define a quantum Hamiltonian map as a map that acts on the star algebra as inner derivations, which match their classical counterpart exactly. We follow the argument here for $X$ and then extend it to study equivariance and covariance of the star product on M. We can show using the $G$-invariance of $\star_X$ and the fact that classical Hamiltonian flows $X_\xi$ of $J(\xi)$ preserve the symplectic structure, that it is a quantum Hamiltonian for the $\mathfrak{g}$-action on $(C^\infty(X)[[h]], \star_X)$:

$$\frac{1}{ih}\,\mathrm{ad}_{\star_X}(J(\xi))(F) := \frac{1}{ih}\big(J(\xi) \star_X F - F \star_X J(\xi)\big) = -\mathcal{L}_{X_\xi} F \tag{24}$$

for all $F \in C^\infty(X)[[h]]$.
We now require that they also be a Lie algebra homomorphism with $\mathfrak{g}$, which we would then turn into a quantum moment map

5.2. **The quantum defect on $X$.** The defect of $J$ from being a Lie algebra homomorphism is

$$\lambda(\xi, \eta) := \frac{1}{ih}\big[J(\xi),\ J(\eta)\big]_{\star_X} - J([\xi, \eta]) \tag{25}$$

**Lemma 5.1.** $\lambda(\xi, \eta) \in \mathbb{C}[[h]]$ *for all* $\xi, \eta \in \mathfrak{g}$

*Proof.* For any $F \in C^\infty(X)[[h]]$, the Jacobi identity for $\star_X$ and equation (24) give

$$\begin{aligned}
\frac{1}{ih}\big[\lambda(\xi,\eta),\ F\big]_{\star_X} &= \frac{1}{(ih)^2}\big[\big[J(\xi),\ J(\eta)\big]_{\star_X},\ F\big]_{\star_X} - \frac{1}{ih}\big[J([\xi,\eta]),\ F\big]_{\star_X} \\
&= \frac{1}{ih}\Big[J(\xi),\ \frac{1}{ih}[J(\eta), F]_{\star_X}\Big]_{\star_X} + \frac{1}{ih}\Big[\frac{1}{ih}[J(\xi), F]_{\star_X},\ J(\eta)\Big]_{\star_X} + \mathcal{L}_{[\xi,\eta]} F \\
&= -\mathcal{L}_\xi \mathcal{L}_\eta F + \mathcal{L}_\eta \mathcal{L}_\xi F + \mathcal{L}_{[\xi,\eta]} F \ = \ 0
\end{aligned}$$

Since $X$ is symplectic, the center of $(C^\infty(X)[[h]], \star_X)$ is $\mathbb{C}[[h]]$ □

By Proposition 4.8 of [8], $\lambda$ is explicitly

$$\lambda(\xi, \eta) = (\omega + \Omega)(X_\xi, X_\eta) - J([\xi, \eta]) \tag{26}$$

also, $[\lambda] \in H^2(\mathfrak{g}, \mathbb{C}[[h]])$, which is a subgroup in the cohomology group of the Chevalley-Eilenberg complex $C(\mathfrak{g}, C^\infty(X)[[h]])$

Since $J([\xi,\eta]) = \omega(X_\xi, X_\eta)$, and $\Omega$ starts at first order in $h$, we have

$$\lambda(\xi,\eta) = \Omega(X_\xi, X_\eta) \tag{27}$$

A *quantum moment map* on $X$ is a correction $J^a(\xi) = J(\xi) - a(\xi)$ with $a \in C^1(\mathfrak{g},\mathbb{C})[[h]]$ satisfying $\lambda = \delta a$, i.e. $\lambda(\xi,\eta) = -a([\xi,\eta])$, giving

$$\frac{1}{ih}\big[J^a(\xi),\, J^a(\eta)\big]_{\star_X} = J^a([\xi,\eta]) \qquad \text{for all } \xi,\eta \in \mathfrak{g} \tag{28}$$

Such $a$ exists if and only if $[\lambda] = 0$ in $H^2(\mathfrak{g}, \mathbb{C}[[h]])$.
For $\Omega = 0$, equation (27) gives $\lambda = 0$, so $a = 0$ and the classical moment map is already quantum, for any $\mathfrak{g}$.
For semisimple $\mathfrak{g}$, the second Whitehead lemma gives $H^2(\mathfrak{g},\mathbb{C}) = 0$, so $\lambda = \delta a$ for a unique $a \in C^1(\mathfrak{g}, \mathbb{C}[[h]])$. Uniqueness of $a$ follows from $H^1(\mathfrak{g},\mathbb{C}) = 0$ (first Whitehead lemma), since the ambiguity in $a$ lies in $Z^1(\mathfrak{g},\mathbb{C}) = 0$
Explicitly, writing $\xi = \sum_k [\zeta^{(k)}, \eta^{(k)}]$ (which is always possible since $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$), the Whitehead homotopy gives

$$a(\xi) = -\sum_k \lambda(\zeta^{(k)}, \eta^{(k)}) = -\sum_k \Omega(X_{\zeta^{(k)}}, X_{\eta^{(k)}}) \tag{29}$$

**Proposition 5.2.** *$\sigma_X^{-1}(J^a(\xi))$ lies in $\ker D_{\mathrm{Fed}} \cap \mathcal{W}^\perp$ for all $\xi \in \mathfrak{g}$.*

*Proof.* We have $\sigma_X^{-1}(J^a(\xi)) = \sigma_X^{-1}(J(\xi)) - a(\xi)\cdot 1 = \Phi(\bar\sigma^{-1}(\rho(\xi))) - a(\xi)\cdot 1$, where 1 is the unit of $\mathcal{W}$. The first term lies in $\ker D_{\mathrm{Fed}} \cap \mathcal{W}^\perp$ by construction (23). The constant section $a(\xi)\cdot 1$ satisfies $D_{\mathrm{Fed}}\, 1 = 0$ and $D^\perp\, 1 = 0$. □

5.3. **The quantum moment map on $M$.** Since $\sigma_X^{-1}(J^a(\xi)) \in \ker D_{\mathrm{Fed}} \cap \mathcal{W}^\perp$, it descends to $M$ via $\bar\sigma \circ \Phi^{-1}$:

$$\hat\rho(\xi) := \bar\sigma\big(\Phi^{-1}(\sigma_X^{-1}(J^a(\xi)))\big) = \rho(\xi) - a(\xi) \tag{30}$$

**Theorem 5.3** (Quantum moment map on Poisson manifolds)**.** *Let $(M,\pi)$ be a Poisson manifold with Hamiltonian $G$-action, classical moment map $\rho : \mathfrak{g} \to C^\infty(M)$, and $G$-invariant connection $\nabla_M$. Let $\Omega$ be $G$-invariant. Then:*

1. *For $\Omega = 0$, the star product $\star_M$ admits the quantum moment map $\hat\rho = \rho$ for any Lie algebra $\mathfrak{g}$:*

$$\frac{1}{ih}\big[\rho(\xi),\, \rho(\eta)\big]_{\star_M} = \rho([\xi,\eta]) \tag{31}$$

2. *For general $\Omega$, the obstruction to the existence of a quantum moment map is $[\lambda] \in H^2(\mathfrak{g},\mathbb{C})[[h]]$ where $\lambda(\xi,\eta) = \Omega(X_\xi, X_\eta)$.*
3. *If $\mathfrak{g}$ is semisimple, then $H^2(\mathfrak{g},\mathbb{C}) = 0$, so the obstruction vanishes for every $G$-invariant $\Omega$. The quantum moment map $\hat\rho(\xi) = \rho(\xi) - a(\xi)$, with $a \in C^1(\mathfrak{g},\mathbb{C})[[h]]$ given by (29), is the unique map satisfying*

$$\frac{1}{ih}\big[\hat\rho(\xi),\, \hat\rho(\eta)\big]_{\star_M} = \hat\rho([\xi,\eta]) \qquad \textit{for all } \xi,\eta \in \mathfrak{g} \tag{32}$$

*Proof.*

1. Equation (27) gives $\lambda = 0$ when $\Omega = 0$, so $a = 0$ and $J^a = J$. The Lie homomorphism property (28) holds on $X$ with no correction.
2. Follows from (27)

(3) $H^2(\mathfrak{g},\mathbb{C})=0$ by the second Whitehead lemma, so $\lambda=\delta a$ with $a$ unique (uniqueness by $H^1(\mathfrak{g},\mathbb{C})=0$).

For the Lie algebra homomorphism property on $M$: the map $\bar{\sigma}\circ\Phi^{-1}:\ker D_{\mathrm{Fed}}\cap\mathcal{W}^{\perp}\to(C^\infty(M)[[h]],\star_M)$ is an algebra homomorphism by Theorem 3.3 and the definition of $\star_M$. Applying it to (28) gives (32).

□

**Remark 5.4.** Fedosov's original construction [5] extends directly to regular Poisson manifolds as explained in the same paper, where the bivector $\pi$ has constant rank, by performing the Weyl bundle construction leafwise along the symplectic foliation. This requires no additional data beyond a Poisson connection along the leaves. The present construction makes no rank assumption on $\pi$. In the regular case both methods produce star products classified by the same Fedosov class $\Omega$, but the groupoid approach additionally provides a natural framework for constructing quantum moment maps, which the leafwise method does not address.

## 6. Obstruction for non-semisimple $\mathfrak{g}$

6.1. **The obstruction cocycle.** For general $\mathfrak{g}$, the obstruction to the existence of a quantum moment map for $\star_M$ with Fedosov class $\Omega$ is the cohomology class

$$[\lambda]\in H^2(\mathfrak{g},\mathbb{C})[[h]] \tag{33}$$

where, by (27), $\lambda(\xi,\eta)=\Omega(X_\xi,X_\eta)$. Writing $\Omega=h\Omega_1+h^2\Omega_2+\cdots$ with each $\Omega_n$ closed,

$$\lambda(\xi,\eta)=h\,\Omega_1(X_\xi,X_\eta)+h^2\,\Omega_2(X_\xi,X_\eta)+\cdots \tag{34}$$

and the obstruction at order $h^n$ is $[\Omega_n(X_\xi,X_\eta)]\in H^2(\mathfrak{g},\mathbb{C})$. If $H^2(\mathfrak{g},\mathbb{C})\neq 0$, this can be nonzero for generic $G$-invariant $\Omega$.

6.2. **Dependence on $\Omega$.** Different $G$-invariant choices of $\Omega$ produce star products with different obstruction classes. By Corollary 4.7 of [8], the condition $i_{X_\xi}\Omega=0$ for all $\xi\in\mathfrak{g}$ implies $\Omega(X_\xi,X_\eta)=0$ and hence $\lambda=0$. This is the condition for the star product to be strongly invariant, in which case the classical moment map is already quantum: $\hat{\rho}=\rho$. The space of $G$-invariant Fedosov classes admitting a quantum moment map is

$$\left\{\Omega\in hH^2_{\mathrm{dR}}(X)^G[[h]]\;\middle|\;[\Omega(X_\xi,X_\eta)]=0\text{ in }H^2(\mathfrak{g},\mathbb{C})[[h]]\right\} \tag{35}$$

For semisimple $\mathfrak{g}$ this is all of $hH^2_{\mathrm{dR}}(X)^G[[h]]$. For non-semisimple $\mathfrak{g}$ with $H^2(\mathfrak{g},\mathbb{C})\neq 0$, the condition (35) is a nontrivial restriction, generalizing the Müller-Bahns–Neumaier condition from symplectic to Poisson manifolds.

**Remark 6.1.** On a symplectic manifold, every star product is equivalent to a Fedosov one, and the equivalence classes are parametrized by $\frac{1}{h}[\omega]+H^2_{\mathrm{dR}}(M,\mathbb{C})[[h]]$. For general Poisson manifolds however, Kontsevich's formality theorem [7] shows that equivalence classes of star products are parametrized by formal deformations of the Poisson structure, classified by the Poisson cohomology $H^2_\pi(M)[[h]]$. The star products constructed here are controlled by the de Rham class $\Omega\in H^2_{\mathrm{dR}}(X)^G[[h]]$ of the ambient formal symplectic manifold. Since the natural map $H^2_{\mathrm{dR}}(X)\to H^2_\pi(M)$ induced by the embedding is in general neither injective nor surjective, the present construction produces a subfamily of all star products on $M$. It does, however, produce precisely those star products for which the question of quantum moment maps can be addressed by the methods of this paper. We conjecture that the equivariant classification of pairs $(\star_M, \hat{\rho})$ by the second equivariant cohomology, described in the symplectic setting by Reichert-Waldmann [9], can also be extended to the Poisson setting from the presented construction by transferring their arguments to the formal symplectic manifold $X$ and descending to $M$ via the algebra isomorphism $\bar{\sigma} \circ \Phi^{-1}$

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